

Owais Siddiqi

✉ omsiddiqi01@gmail.com | 💻 owaissiddiqi.co.uk | in [Owais Siddiqi](#) | 🌐 [Owais Siddiqi](#) | 📍 London, UK

Professional Summary

Computational Biologist and Bioinformatics Data Scientist with three years of industry experience analysing large-scale genomic and transcriptomic datasets and building predictive models for patient stratification in oncology. Proven track record of translating multi-omics and clinical data into deployable diagnostic tools, from RNA-seq workflow design and ML model validation through to production ETL pipelines embedded in R&D workflows. Experienced collaborating across scientific, engineering, and clinical teams in fast-paced, purpose-led environments.

Work Experience

CellBxHealth plc

Jan 2025 – Mar 2026

Data Scientist

- **Clinical Data Engineering and ETL:** Developed automated ETL pipelines integrating imaging, NGS, and clinical metadata, reducing onboarding and QC processing time by 33% per study.
- **NGS Bioinformatics:** Designed and implemented a novel RNA-seq workflow for rare-cell transcriptomic analysis, leading experimental design, QC benchmarking, biological interpretation, and scalability validation. Reduced per-cohort processing time from 6 to 4 working days and secured £150k in internal funding to co-develop a commercial product.
- **Workflow Automation and Operational Enablement:** Automated manual R&D processes and introduced structured SOPs, eliminating approximately 8 hours of manual processing time per week across the team through workflow automation.
- **Machine Learning for Patient Stratification:** Designed, trained, and validated predictive models using real-world molecular and clinical datasets for patient stratification. Models were adopted into scientist workflows for downstream analysis.
- **Integrated Multi-Omics Analytics:** Merged imaging-derived and molecular features to characterise CTC phenotypes and gene-expression programmes, supporting biomarker identification and patient stratification studies.
- **Cross-Functional Communication:** Presented complex analyses to technical teams, non-technical stakeholders, and senior leadership, translating computational findings into actionable insights that aligned R&D, commercial, and executive priorities.

Genevation Ltd.

Apr 2024 – Dec 2024

Junior Genomic Data Scientist

- **AI-Driven Biomarker Discovery:** Designed, trained, and validated XGBoost classification models on tabular VCF-derived somatic variant data to support personalised oncology vaccine development.
- **NGS Workflow Automation:** Engineered containerised pipelines converting FASTQ to VCF, reducing average analysis turnaround from 5 to 2 days per sample batch.
- **RNA-Seq Variant Analysis:** Executed RNA-seq alignment and variant-calling workflows on GCP and HPC environments, delivering reproducible genomic outputs for internal R&D use.
- **Data-Driven Fundraising Strategy:** Produced analytical outputs and visualisations incorporated into VC pitch decks that supported £400k in fundraising.

PDUK Ltd.

Aug 2023 – Jan 2024

Data Scientist Intern

- **Data Engineering and ETL:** Built SQL-based cleansing and transformation workflows across a large customer database, reducing batch processing time by 25%.
- **KPI Dashboards for Senior Leadership:** Developed interactive performance dashboards surfacing revenue, demand, and conversion metrics for the Senior Leadership Team, enabling data-driven commercial planning and weekly board reporting.
- **Forecasting and Predictive Modelling:** Designed and deployed a demand forecasting model, contributing to approximately £18,000 reduction in excess inventory over H2 2023.
- **Customer Segmentation:** Applied k-means clustering to purchase history data to identify high-value customer segments within a B2B ecommerce context, enabling targeted commercial outreach and contributing to approximately £12,000 increase in campaign-driven revenue over six months.

Technical Skills

- **Programming and Data:** Python, R, SQL, Bash, MATLAB, Git; Data wrangling and analytics using Pandas, NumPy, and Jupyter.
- **Machine Learning and AI:** Scikit-learn, LightGBM, TensorFlow, PyTorch. Predictive modelling, clustering using HDBSCAN and GMM, anomaly detection, and survival analysis including Cox PH, time-varying Cox, and Kaplan-Meier.
- **Bioinformatics and Genomics:** Bioinformatics workflows covering QC, alignment, differential expression, fusion detection, and variant calling using STAR-Fusion, Salmon, DESeq2, EdgeR, and Nextflow.
- **Data Visualisation and Reporting:** Streamlit, Plotly, Matplotlib, Seaborn, Excel. Interactive dashboards and reporting for biological and clinical datasets.
- **Cloud and Workflow Automation:** Google Cloud Platform, Docker, CI/CD, Linux and HPC.

Research Experience

- **Owais Siddiqi**, Winfield J., Zormpas-Petridis K., Harris E., et al. *Investigating the Repeatability of Multifrequency Magnetic Resonance Elastography Applied to a Soft Gelatin Phantom*. **31st ISMRM Conference & Exhibition**, London, UK, 2022.
- Ciccioli M., Spode M., Coates M., Duhig E., **Owais Siddiqi**, et al. *Unlocking the Potential of Liquid Biopsy in Glioblastoma: An Extensive Evaluation of Circulating Biomarkers*. **SNO Annual Meeting**, 2025.
- Mezni A., Goodwin C., Greaves D., **Owais Siddiqi**, et al. *The Value of Characterising Circulating Tumour Cells Using Epithelial and Mesenchymal Markers in Breast Cancer*. **7th ACTC Conference**, 2025.
- Geraldizo M.C., Coates M., Torkzadeh F., **Owais Siddiqi**, et al. *Custom Gene Expression Profiling of Circulating Tumour Cells via Amplitude-Based Multiplex Digital PCR*. **7th ACTC Conference**, 2025.

Education

Imperial College London

2022 – 2023

MSc (*Honours*) Biomedical Engineering, Merit

- **CNN Regression Modelling:** Designed and trained a TensorFlow/Keras CNN to predict cell differentiation from microscopy images for phenotypic screening, achieving an R^2 of 0.80. Applied 5-fold cross-validation with hyperparameter tuning, and regularised against overfitting using dropout, L1/L2 penalties, and data augmentation.

Queen Mary University of London

2019 – 2022

BEng (*Honours*) Biomedical Engineering, First Class

- **BCI Classification Project:** Built a multi-class EEG-SSVEP classifier in Python and MATLAB, implementing a bagging ensemble of ANFIS binary classifiers using OAO decomposition and average voting aggregation.